

ESTIMATION OF TEMPERATURE FLUCTUATIONS HARSHNESS REGARDING STABILITY OF STRUCTURES IN THE NANOMETER RANGE

MEDSI 2016, Barcelona

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SYNCHROTRON SOLEIL**



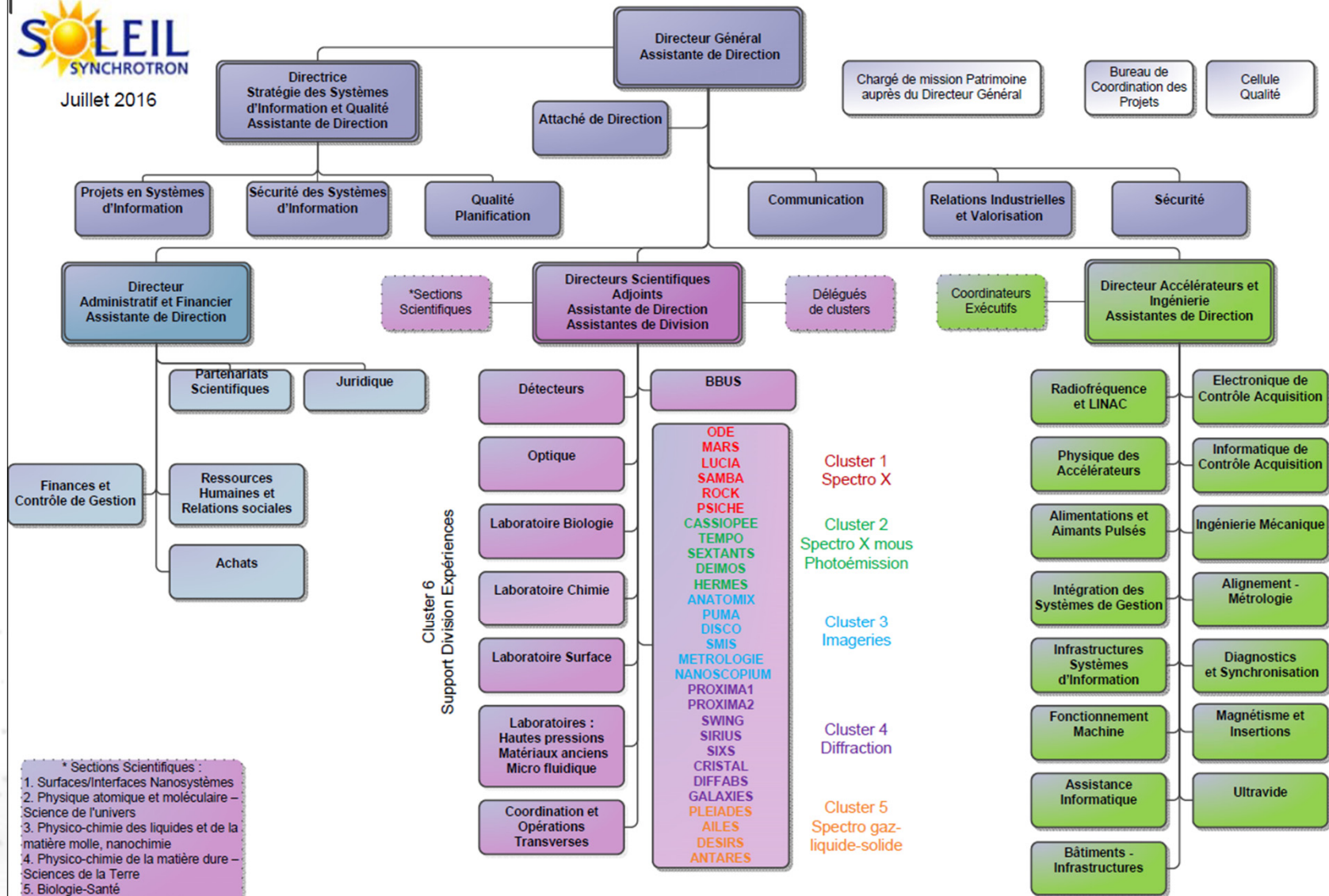
OUTLINE

- **MECHANICAL ENGINEERING AT SOLEIL**
- **NANOMETER STABILITY: SOME CLARIFICATIONS**
- **DISTURBANCE SOURCES CLASSIFICATION**
- **THERMAL STABILITY IN NUMBERS**
- **THERMAL-MECHANICAL SUSCEPTIBILITY**
 - **MECHANISMS AT PLAY**
 - **PROPOSED PRACTICAL RULES FOR ESTIMATION**
 - **ILLUSTRATION AND VALIDATION**
 - **RELIABILITY ANALYSIS**
- **SUMMARY / CONCLUSION**

Mechanical Engineering at SOLEIL



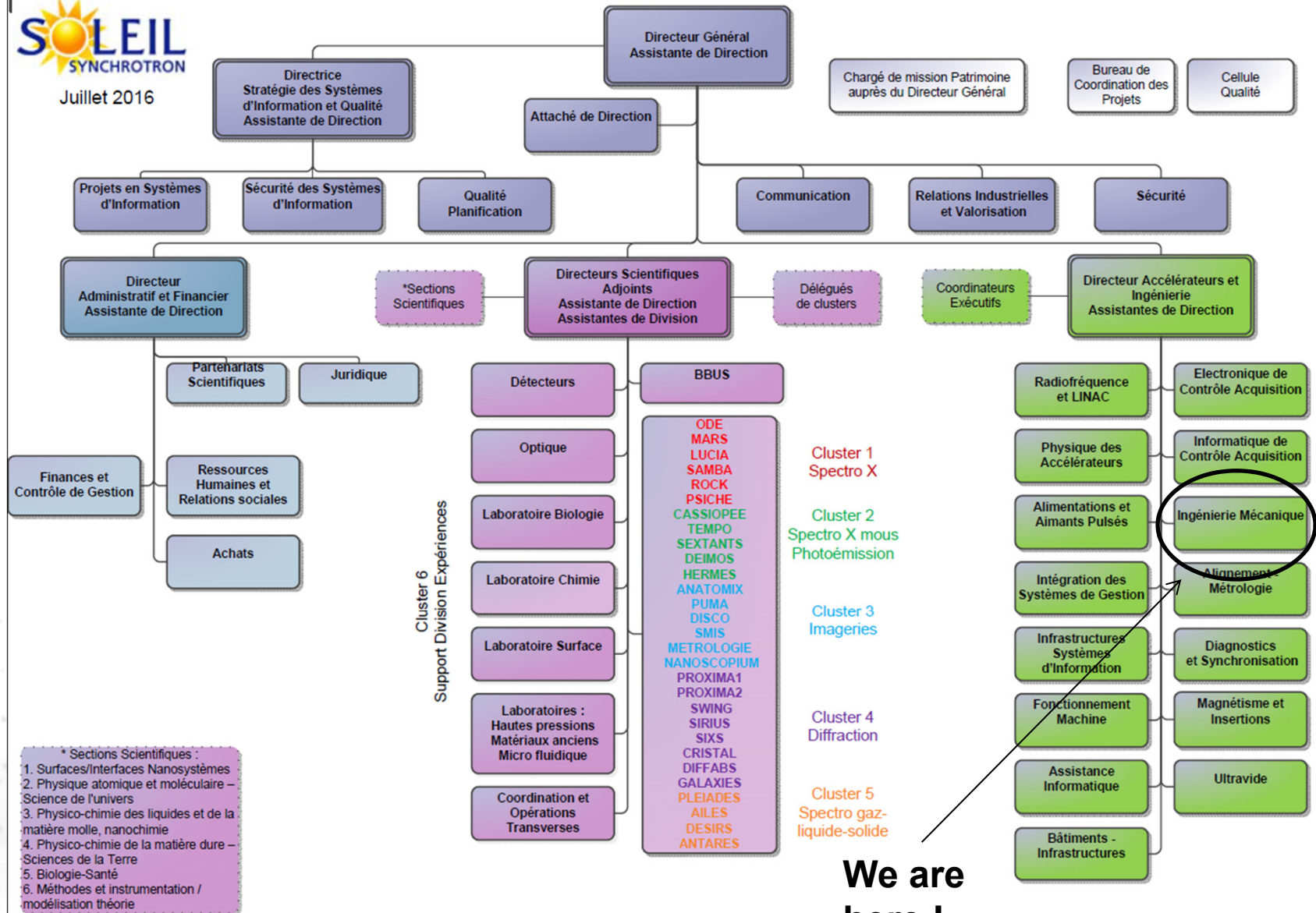
Juillet 2016



Mechanical Engineering at SOLEIL



Juillet 2016

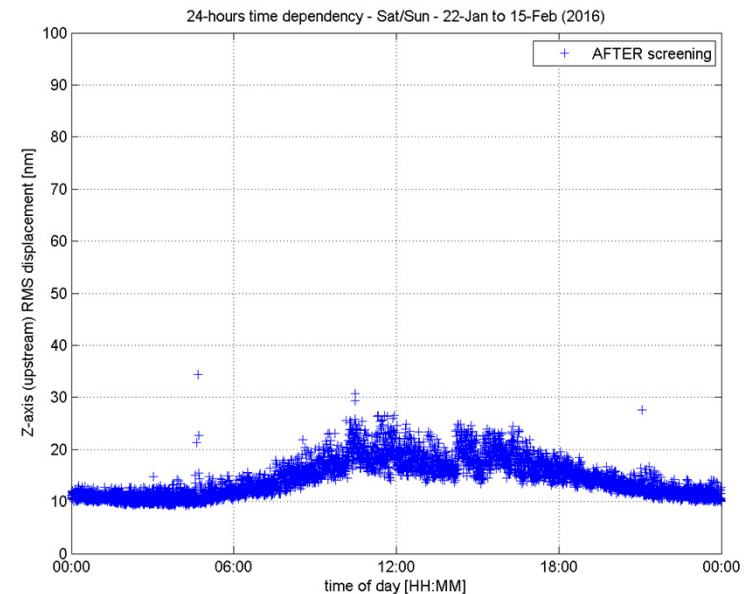
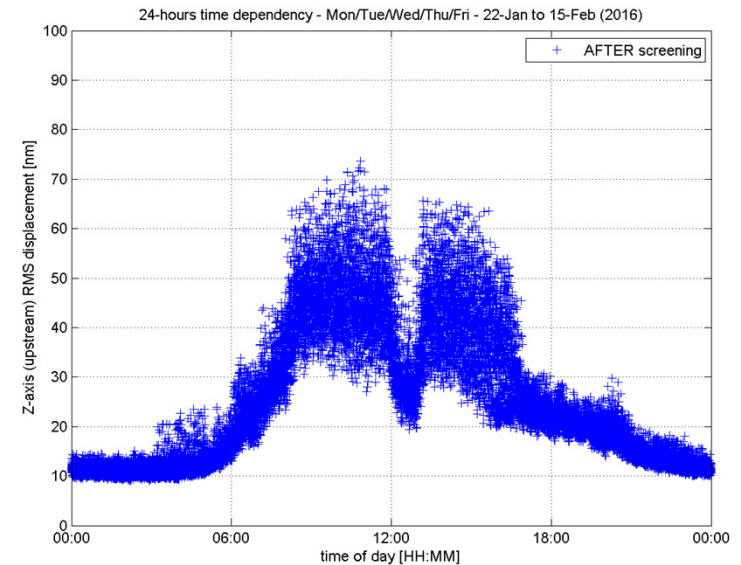


We are here !



NANOMETER STABILITY: SOME CLARIFICATIONS

- Ground motion at SOLEIL = *dozens* of nm (RMS !)
- Absolute stability is not required, but
- Mis-alignment is critical
- Alignment preserved if:
 - Ground motion is uniform
 - Supports deflections are small

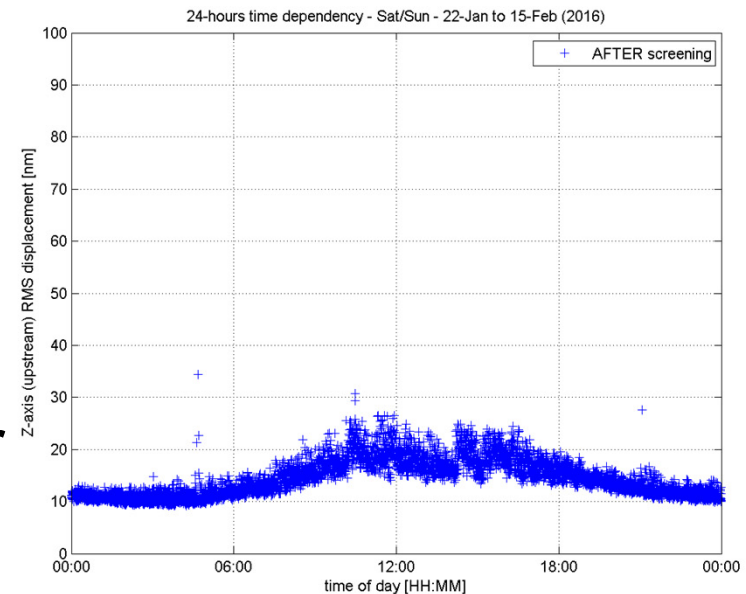
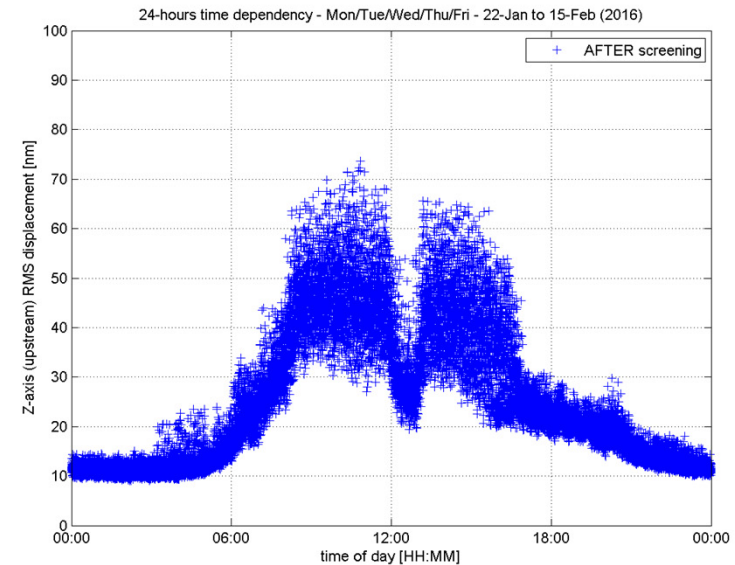


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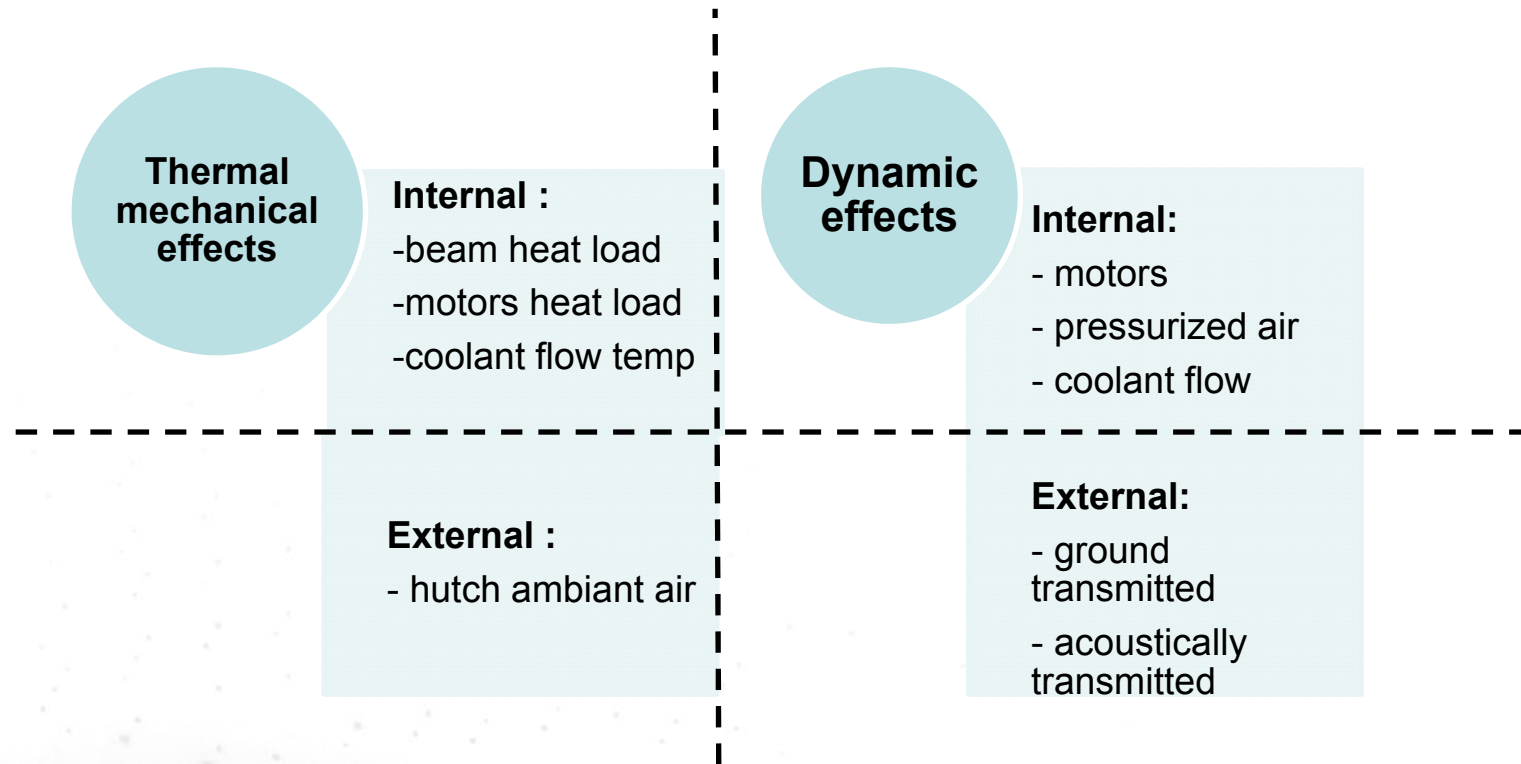
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**Compact & stiff support structures =>
dynamic relative motion $\ll 10\text{nm}$**

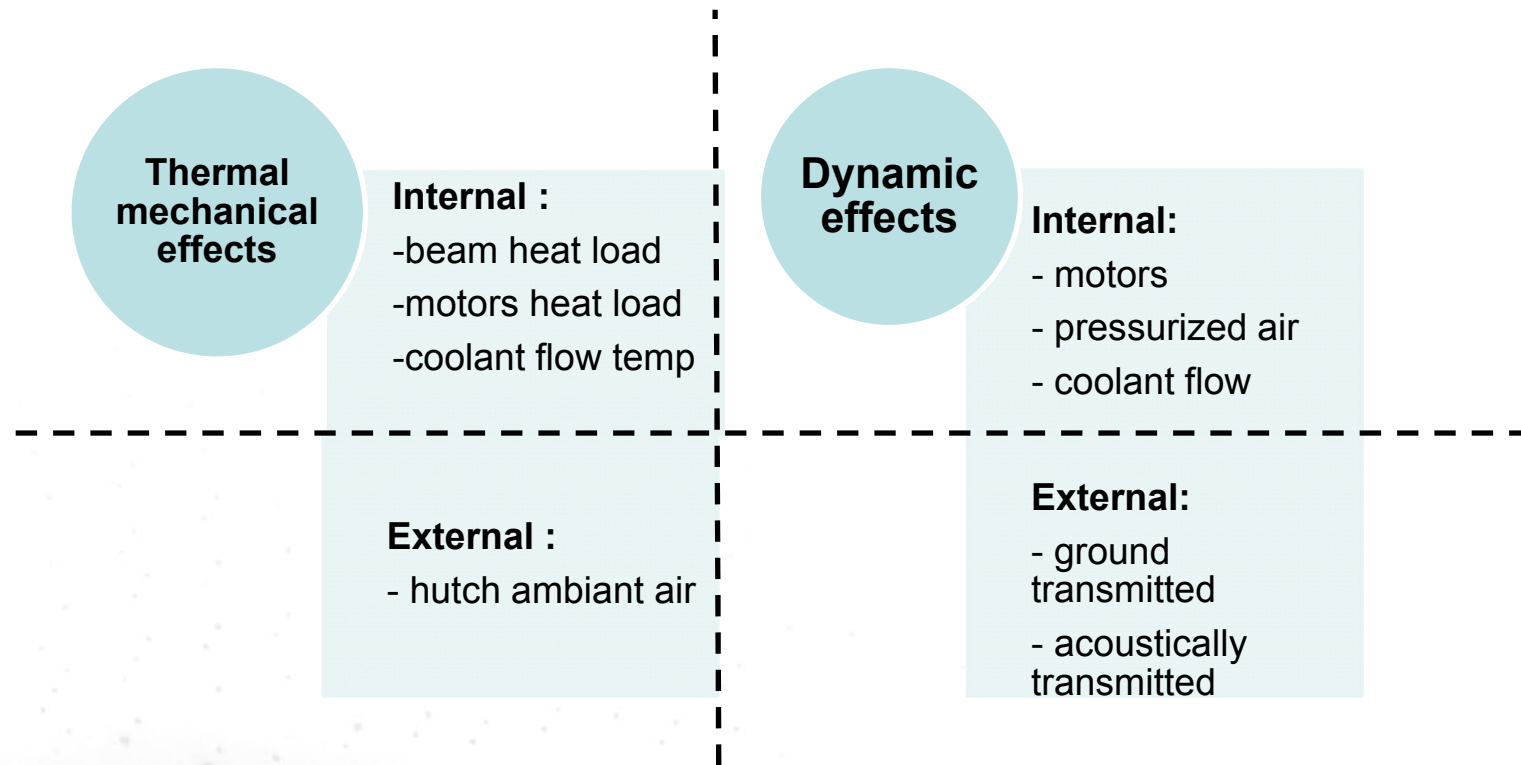
**BUT for long (1h+) observation time,
thermally induced error can be WAY higher**



DISTURBANCE SOURCES CLASSIFICATION



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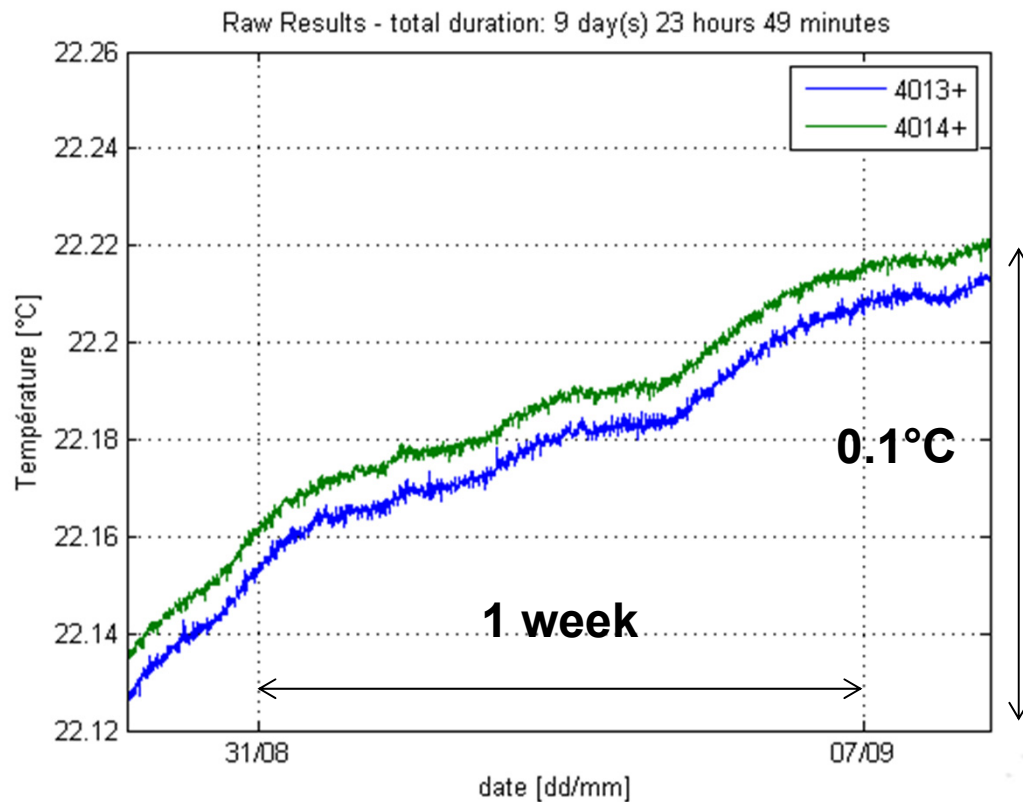


Design: no disturbance from heat sink

Reality: no perfect heat sink, non-zero perturbations

THERMAL STABILITY IN NUMBERS (1/2)

**Common practice :
use peak-to-peak values**

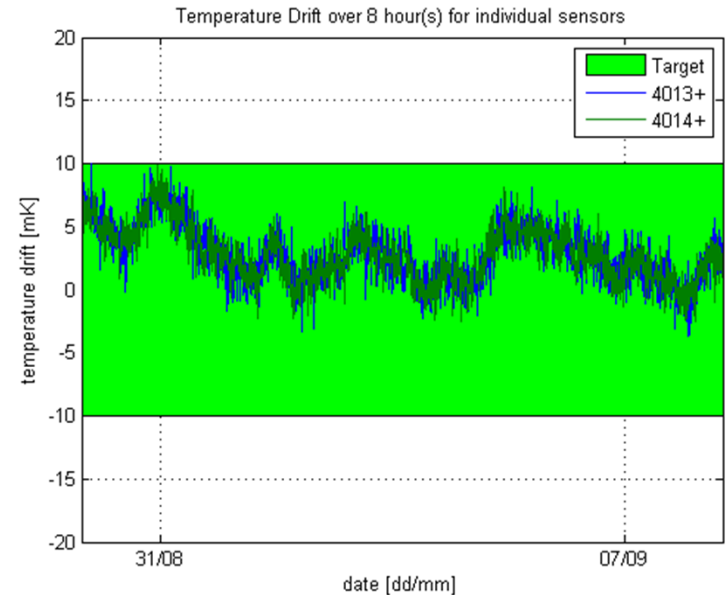
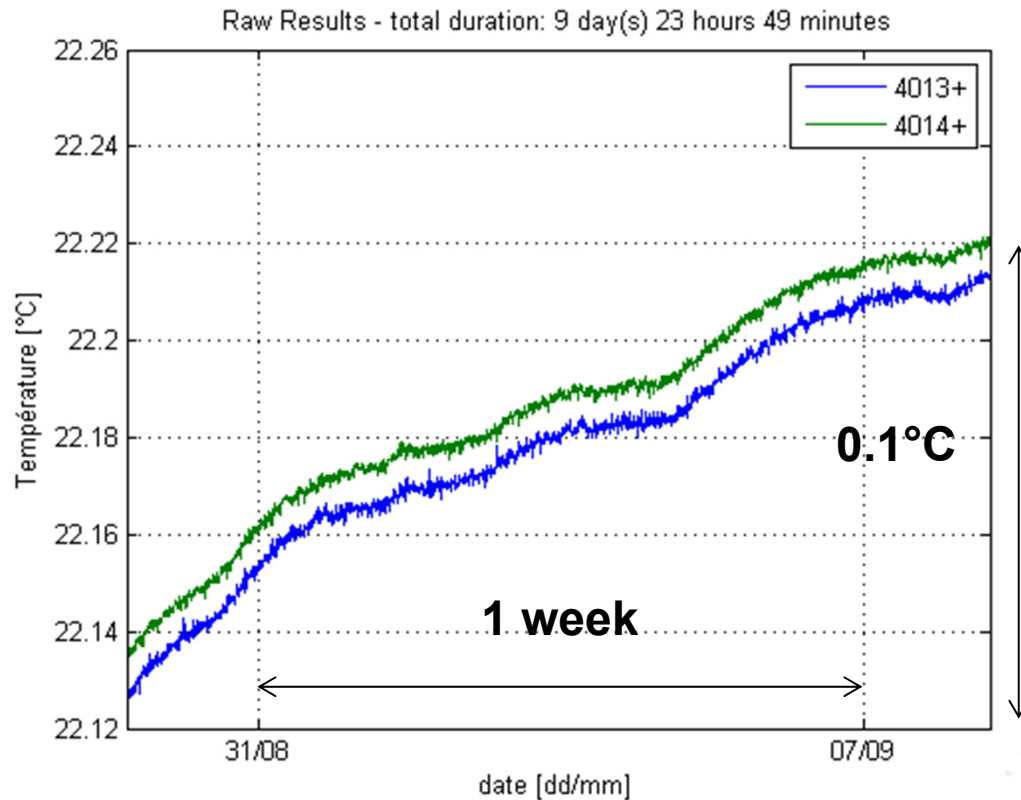


Water buffer - blue=bottom green=top

THERMAL STABILITY IN NUMBERS

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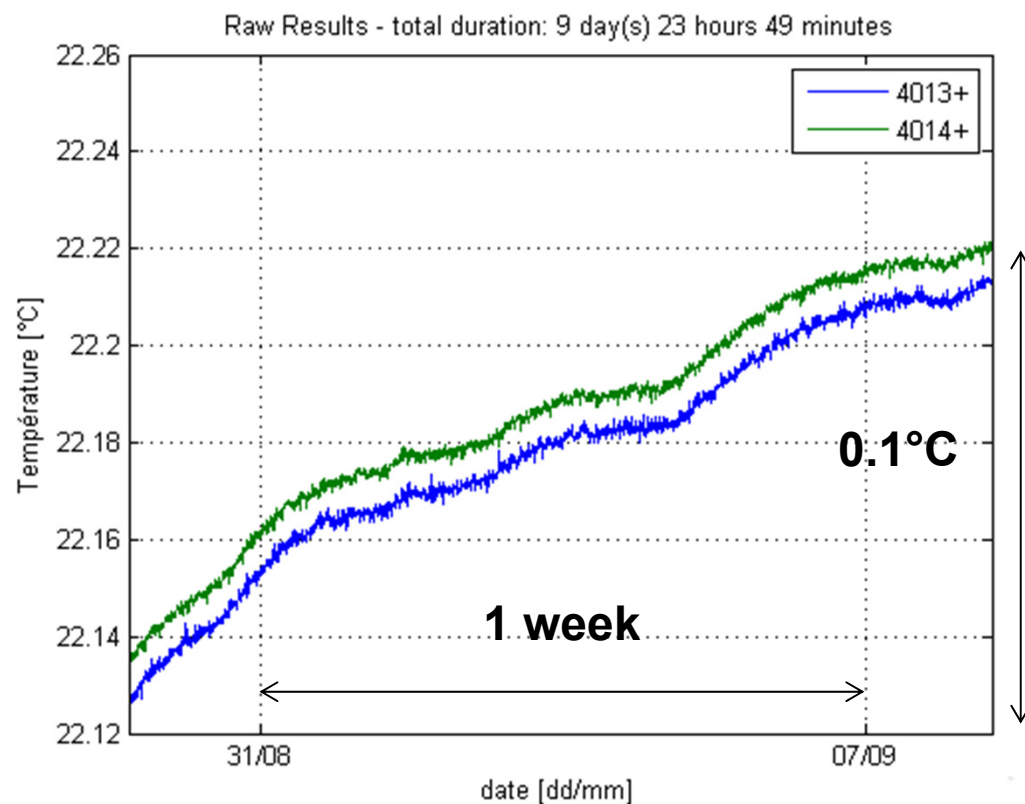


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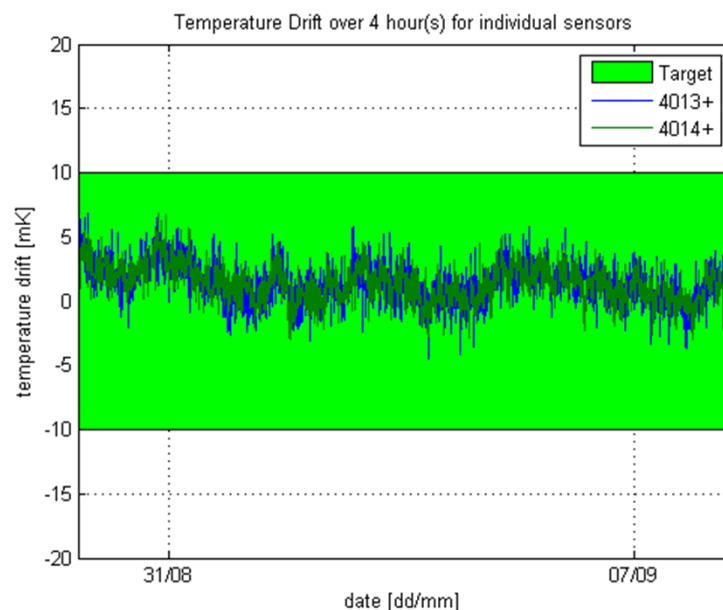
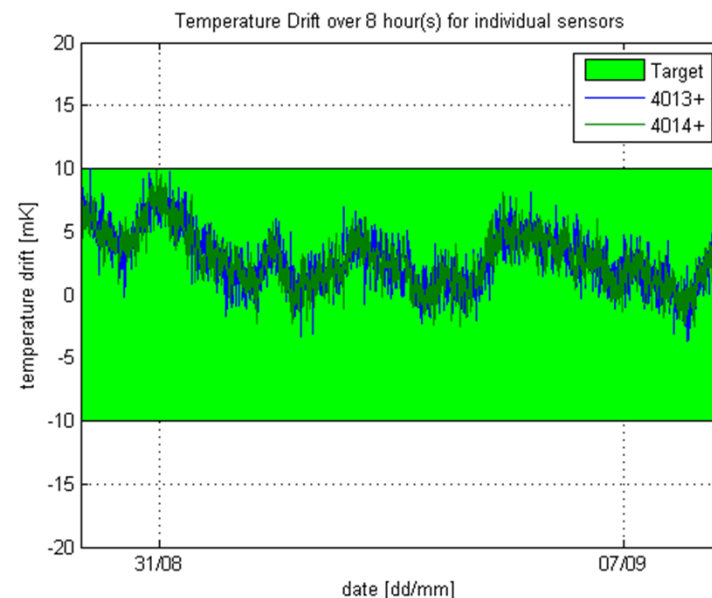
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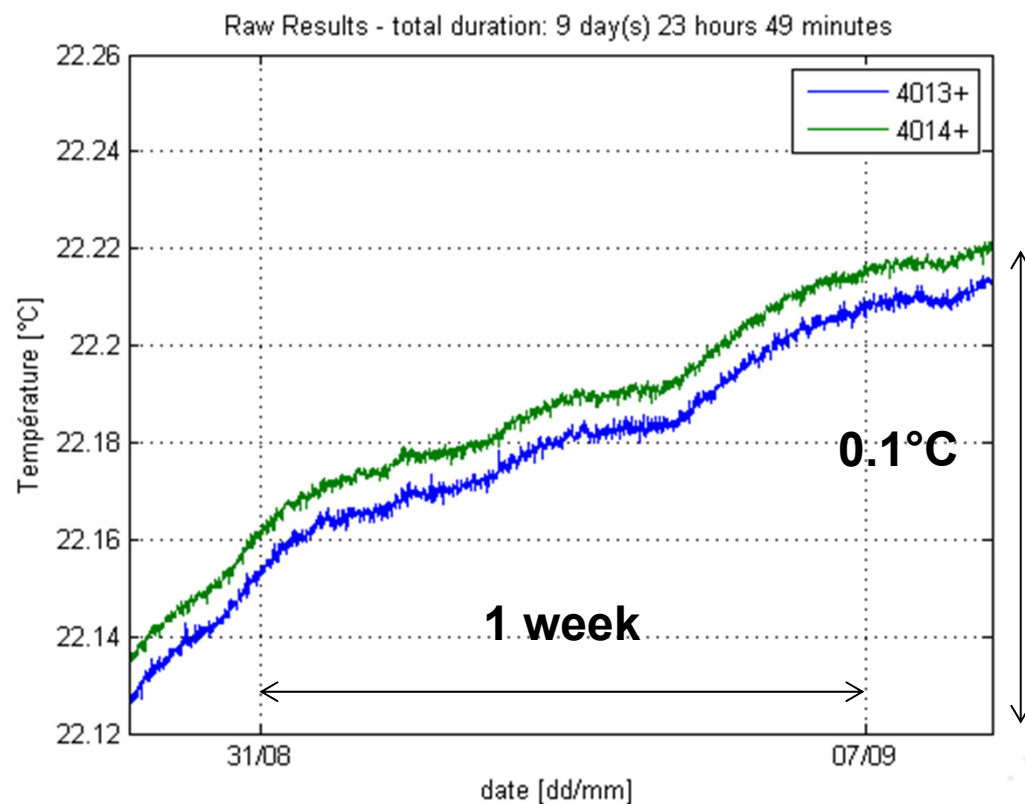
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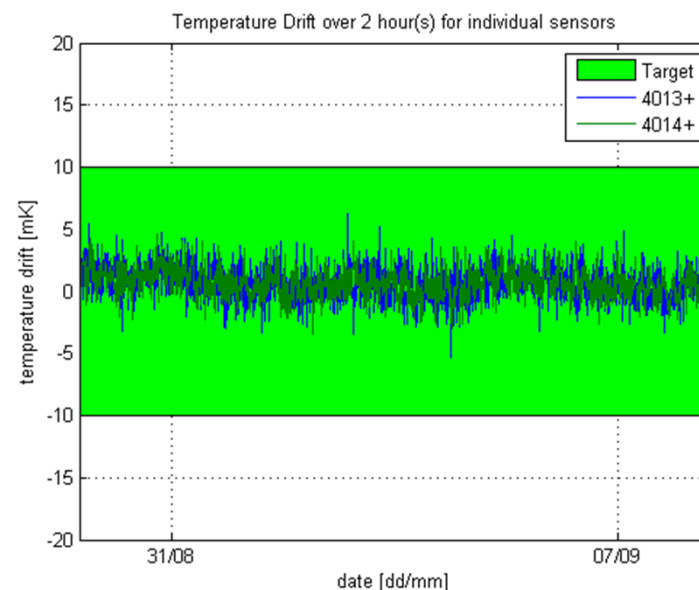
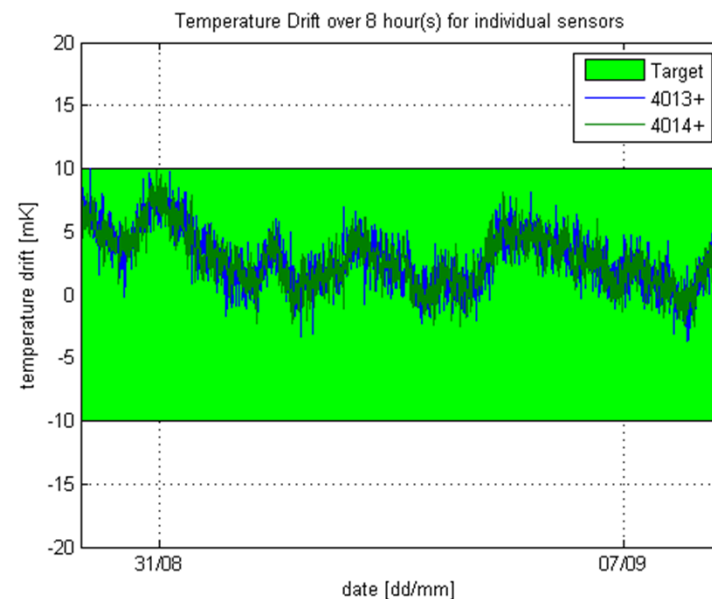
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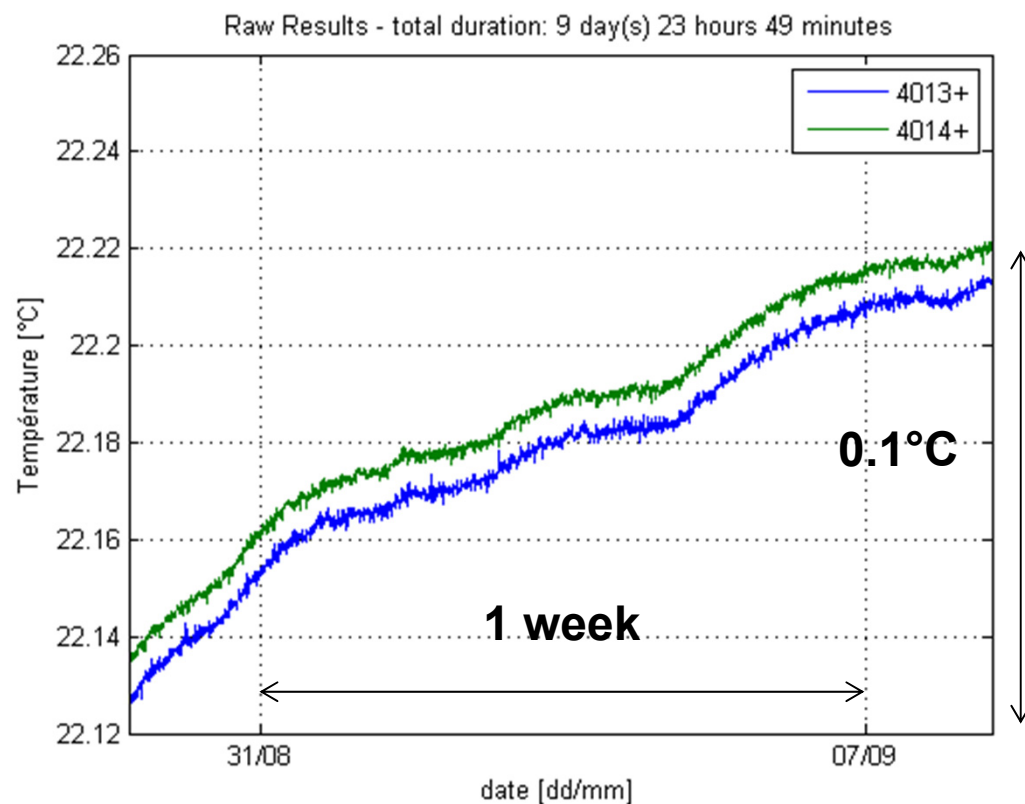
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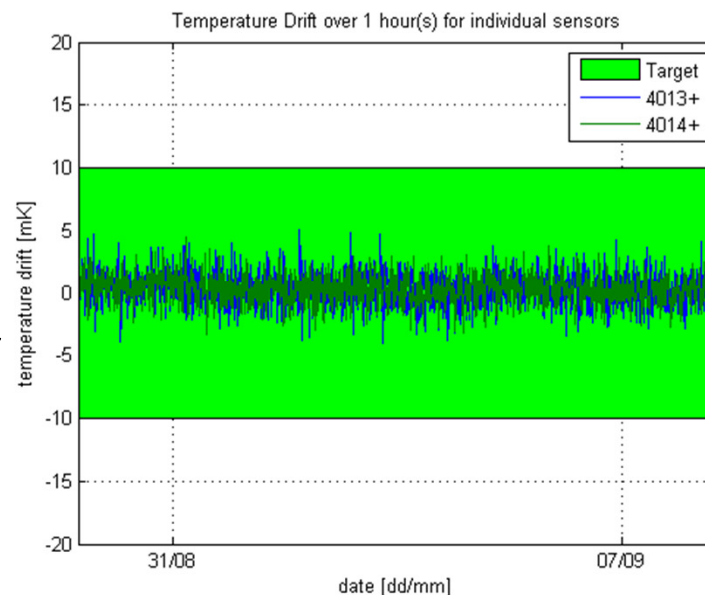
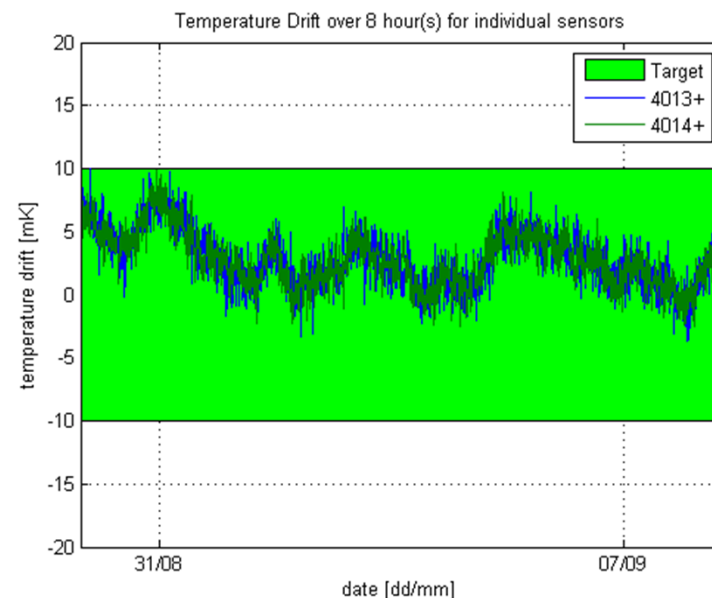
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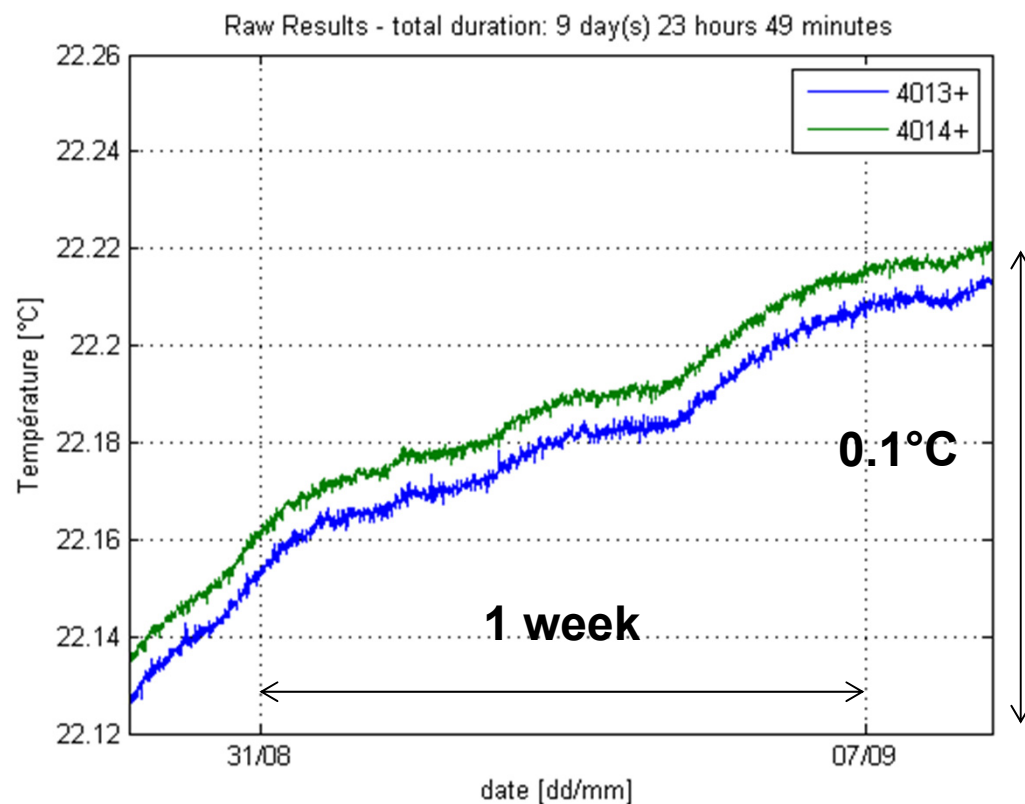
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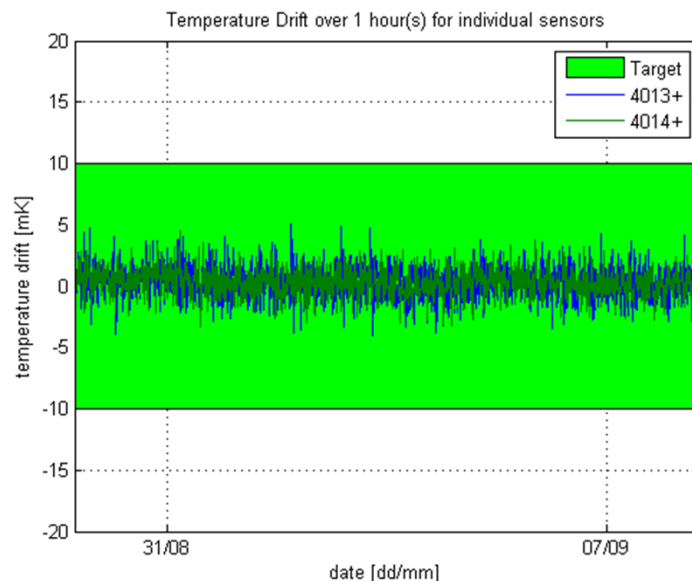
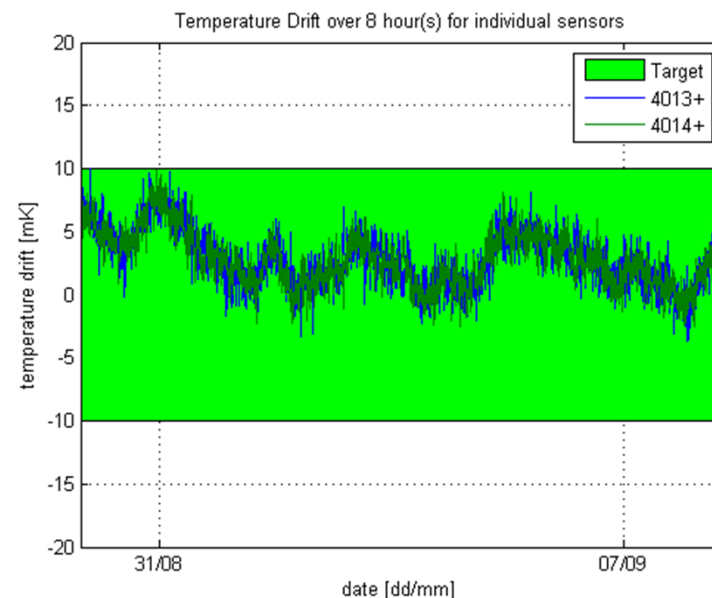
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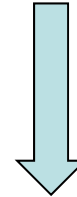
(2/2)

AMPLITUDE PP (millidegree C)

9 days => 100mdeg
8 hours => 12 mdeg
4 hours => 10 mdeg
2 hours => 10 mdeg
1 hour => 10 mdeg

SIGNAL TYPE

Steady trend



No trend (random)

THERMAL STABILITY IN NUMBERS

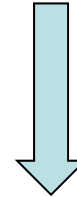
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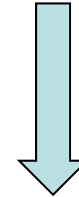
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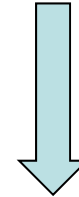
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Q: Where do we stand
actually ?

**A1: our reference fluctuations are 50%
of total error budget, it's a mess !**

**A2: random temperature
fluctuations have no effect
whatsoever so stop worrying:
We're good !**

THERMAL-MECHANICAL SUSCEPTIBILITY : MECHANISMS

Intuition: « fast »
temperature oscillation
do not propagate very
far along support
structure

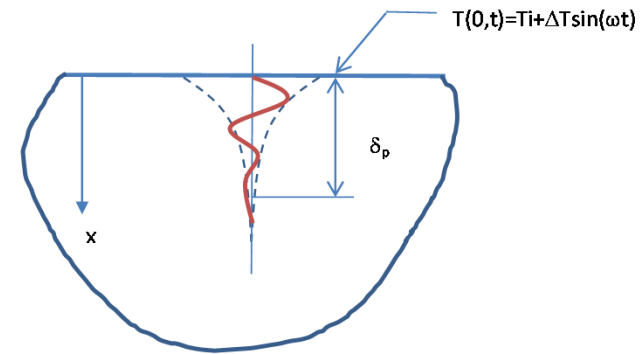
$\alpha = \text{thermal diffusivity} [Length^2/Time]$

$$\alpha \left[\frac{mm^2}{s} \right] \sim 4(Invvar) \dots 100(Al)$$

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Analytical solution:
exponential decay



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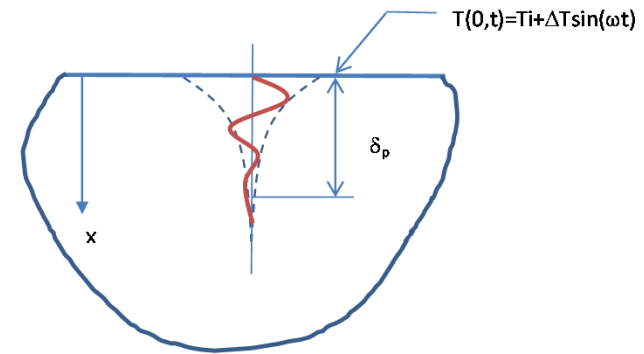
$$\alpha \left[\frac{\text{mm}^2}{\text{s}} \right] \sim 4(\text{Invar}) \dots 100(\text{Al})$$

$$\frac{T(x,t) - T_i}{\Delta T} = \exp\left(-\frac{x}{\sqrt{\frac{\omega}{2\alpha}}}\right) \sin\left(\omega t - x \sqrt{\frac{\omega}{2\alpha}}\right)$$

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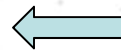


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$$T_{\text{corner}}[\text{s}] \sim 6.3 L[\text{mm}]^2/\alpha$$



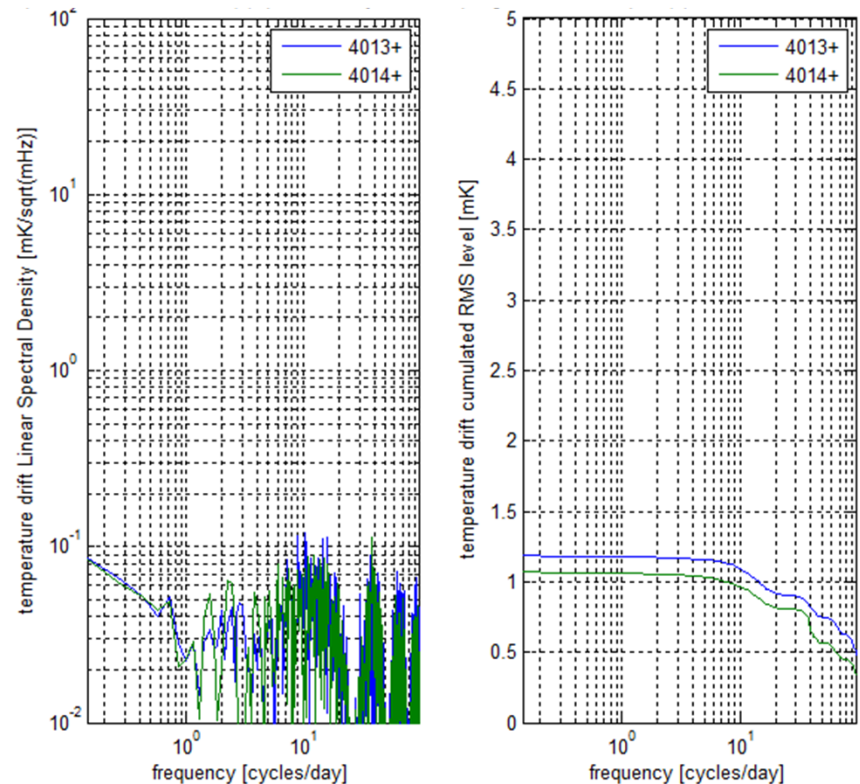
$$\delta_{\text{effective}} = \sqrt{\frac{\alpha \cdot T}{2\pi}}$$

THERMAL-MECHANICAL SUSCEPTIBILITY : PRACTICAL ESTIMATION RULES (1/2)

- Random fluctuations $\Leftrightarrow$ broadband spectral content
- Hence, harmonic formula not usable « as is »
- Need to SEPARATELY estimate severity of ALL frequency components
- Combine individual components into OVERALL harshness indicator

**Need for a simple, automated
procedure**

EXAMPLE : SPECTRAL ANALYSIS OF 8 HOURS TEMPERATURE DRIFT



THERMAL-MECHANICAL SUSCEPTIBILITY : PRACTICAL ESTIMATION RULES (2/2)

Estimate thermal
mechanical « frequency
response function »

$$H_{UT_c}(f)$$

(structural response / unit
temperature fluctuation)

Break down coolant
temperature fluctuations into
individual frequency
components, i.e.

« PSD analysis » $\hat{\Phi}_{T_c T_c}(f)$

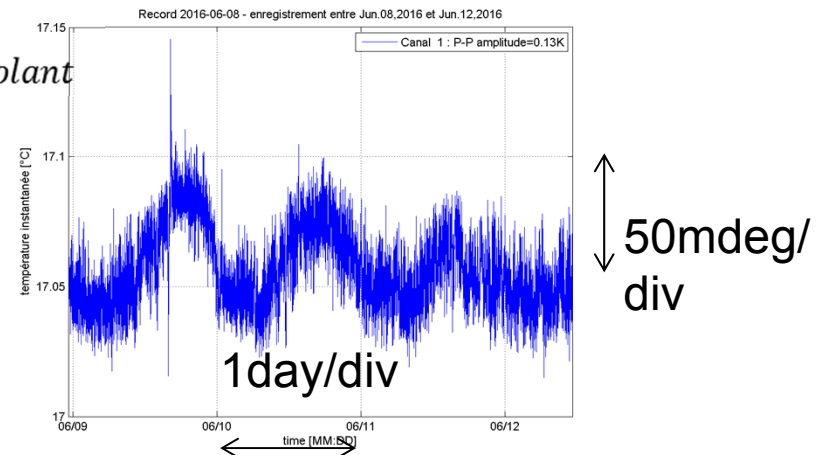
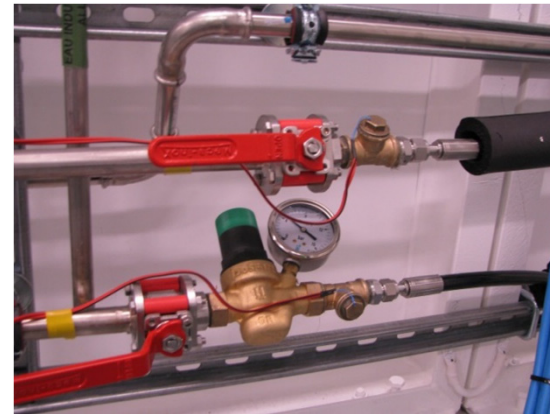
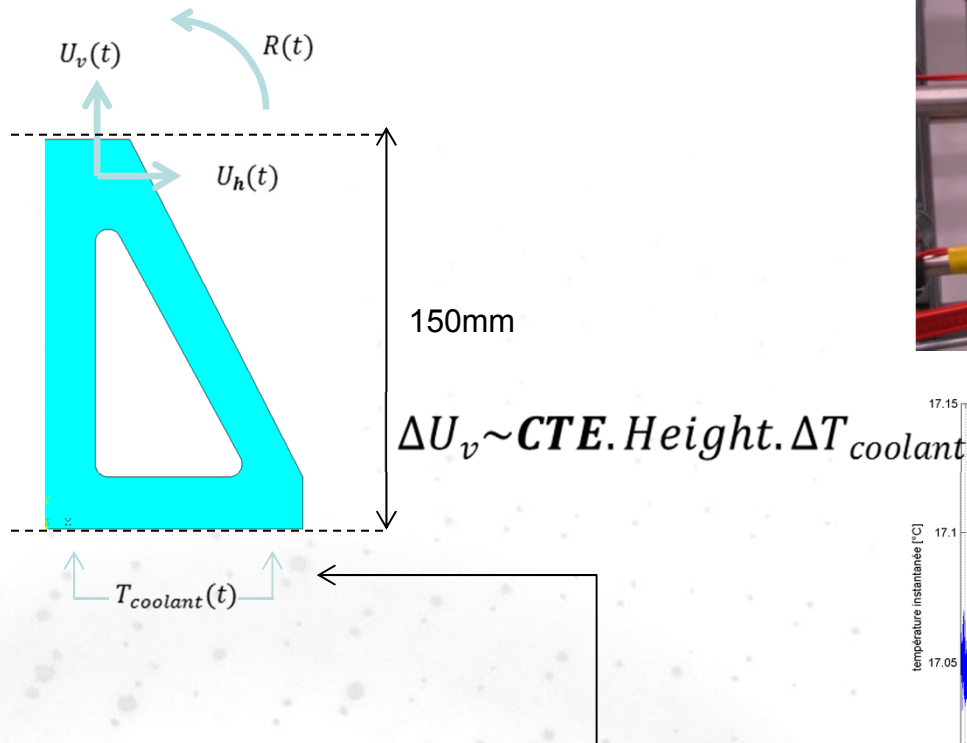
(degree²/unit frequency
bandwidth)

Combine to obtain into actual
response level

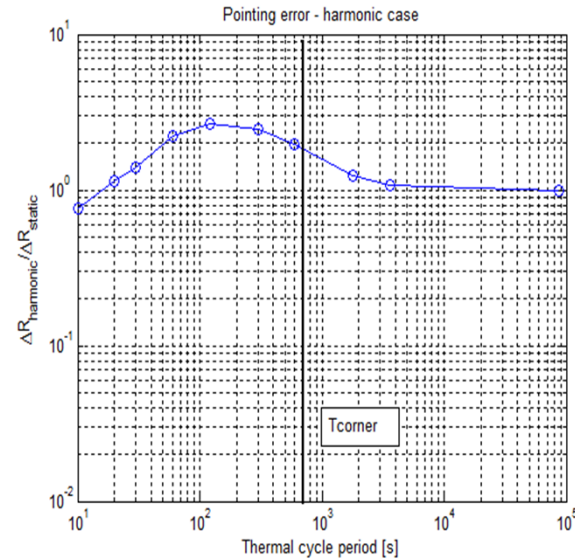
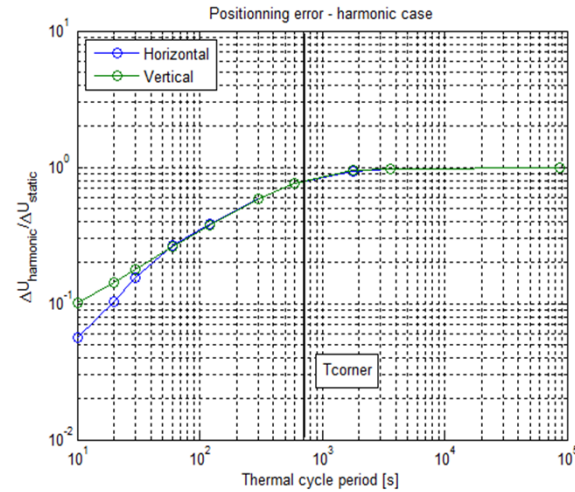
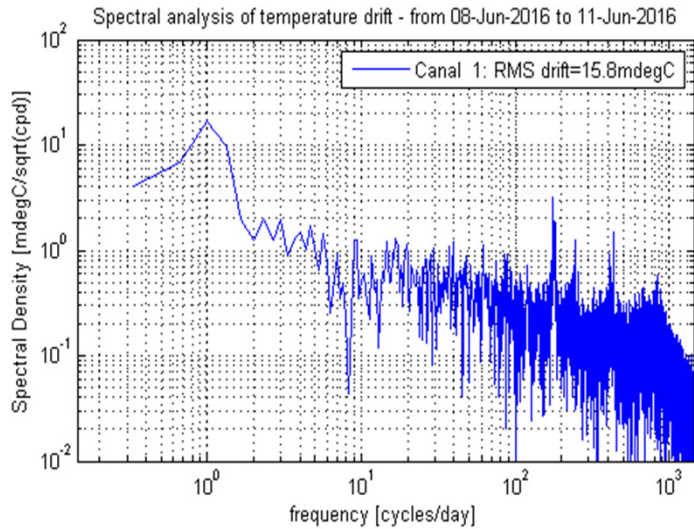
$$U_{rms} = \left(\int_0^{f_{maxi}} |H_{UT_c}(f)|^2 \cdot \hat{\Phi}_{T_c T_c}(f) df \right)^{1/2}$$

THERMAL-MECHANICAL SUSCEPTIBILITY : APPLICATION (1/2)

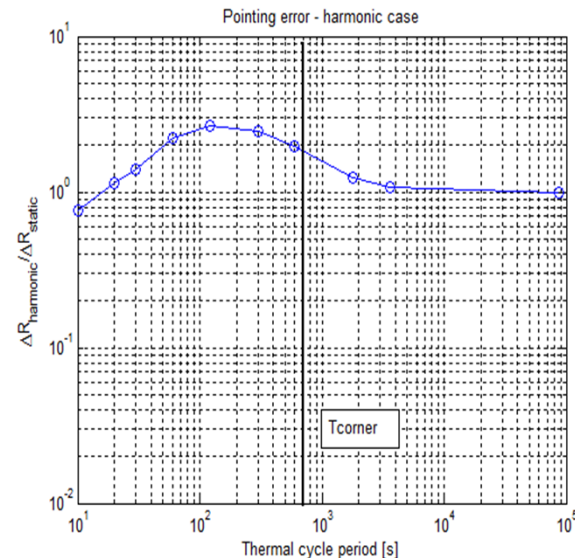
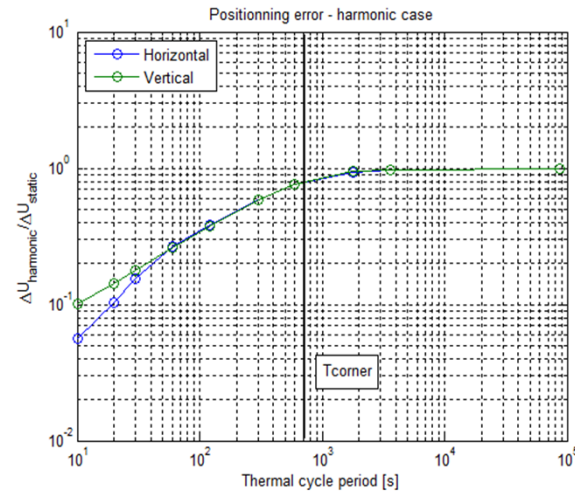
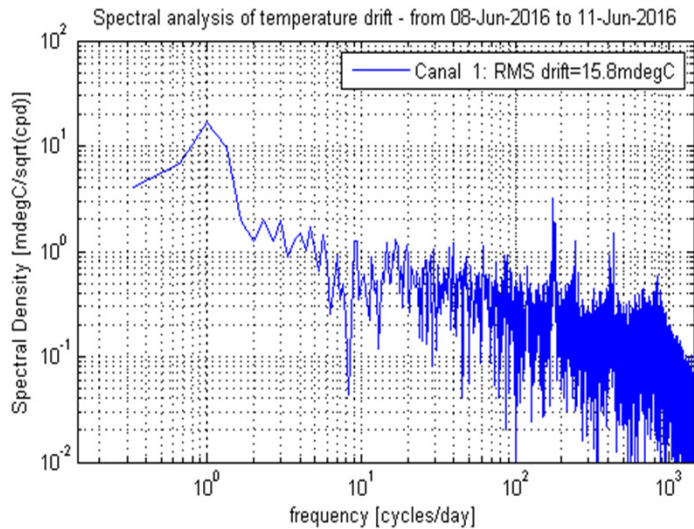
Application case: Aluminum (CTE=23ppm/K, diffusivity=100mm²/s)



THERMAL-MECHANICAL SUSCEPTIBILITY : APPLICATION (2/2)



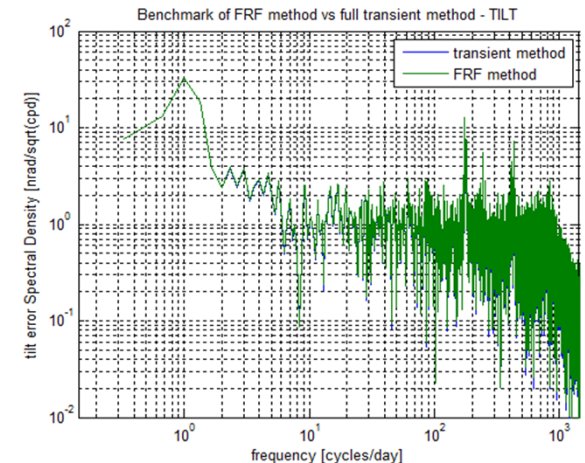
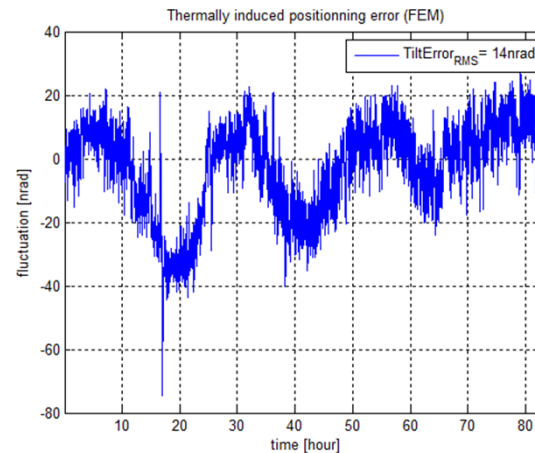
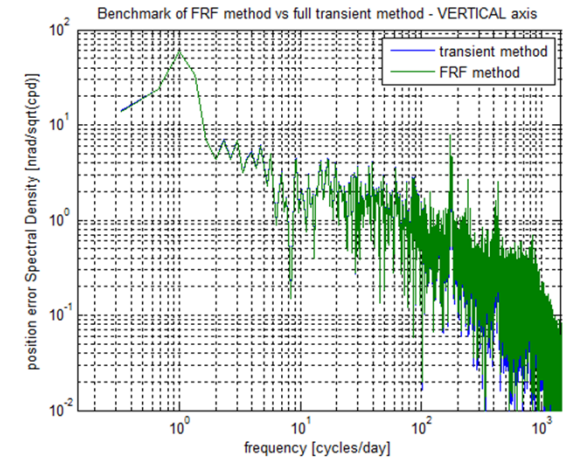
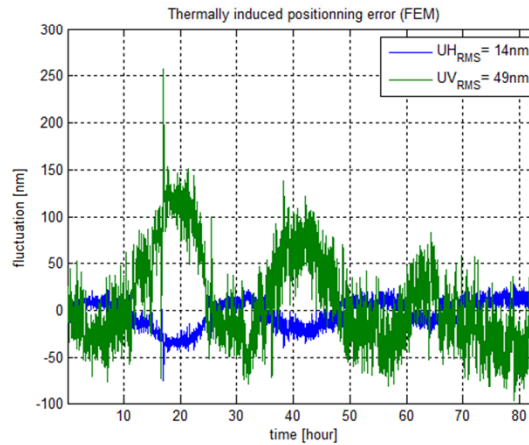
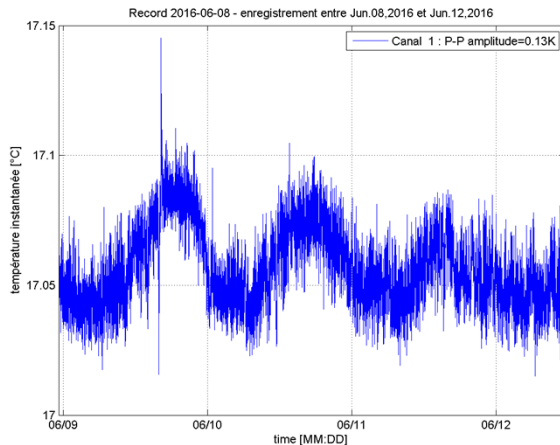
THERMAL-MECHANICAL SUSCEPTIBILITY : APPLICATION (2/2)



Corner period as predicted by hand calc's matches FEM analysis results ☺ ... except for pointing errors ☹ where AMPLIFICATION occurs

THERMAL-MECHANICAL SUSCEPTIBILITY : VALIDATION

- Benchmark vs direct calculations



Near perfect agreement

	Transient	FRF	Ratio
$U_{\text{horizontal}}$	15nm	14nm	0.99
U_{vertical}	50nm	49nm	0.99
$R_{\text{transverse}}$	43nrad	44nrad	1.02

PRACTICAL USAGE: RELIABILITY ANALYSIS

RANDOM FLUCTUATIONS

- Deterministic figures of merit (« Peak to peak ») amplitude meaningless
- Need to define amplitude into a probabilistic sense
- Common practice at SOLEIL: Probability Of Exceedance For Given Allowable Drift (AD)

Need to know:

- RMS amplitude U_{RMS}
- Observation time T_s

$$P(|Drift(T_s)| > AD) \\ = 1 - \operatorname{erf}\left(\frac{AD}{\sqrt{2}U_{RMS}(T_s)}\right)$$

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NB: Other statistics (mean time between exceedances, etc..) could be estimated

SUMMARY / CONCLUSION

For unstable environments, clear trends exist:

- « static rules » apply

BUT, for stabilized environments, fluctuations are mostly random

- need for *ad hoc* estimations procedures

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- When dealing with random phenomena, go ahead, don't be afraid of using statistical indicators 😊

Severity of random temperature fluctuations cannot readily be assessed at face value.

Using the proposed procedure, a minimum analytical effort can provide reliable quantitative estimates of setup reliability

THANK YOU FOR YOUR ATTENTION

QUESTION & COMMENTS ARE
WELCOME

